BioSys: Efficient Quality Control System for Manufacturing of Sustainable Biopolymer Composites

Barney H. Miao*
barneym@stanford.edu
Stanford University
Stanford, California, USA

Yiwen Dong
Stanford University
Stanford, California, USA
ywdong@stanford.edu

Andreas Theissler
Aalen University of Applied Sciences
Aalen, Germany

Andrew Lesh
Stanford University
Stanford, California, USA

David J. Loftus
NASA Ames Research Center
Moffett Field, California, USA

Michael D. Lepech
Stanford University
Stanford, California, USA
mlepech@stanford.edu

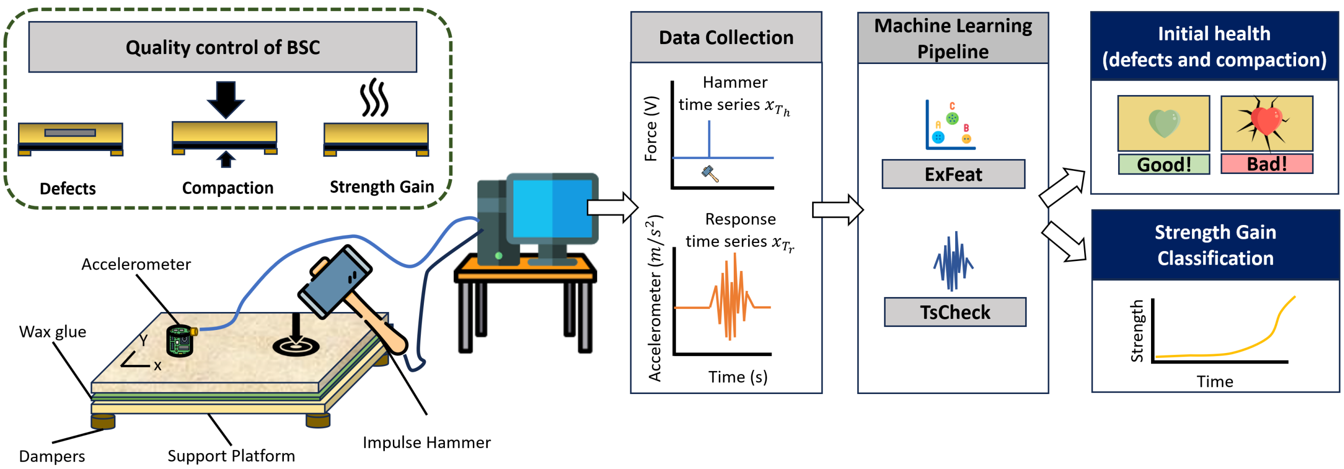

Figure 1: BioSys, a novel quality control system, detects defects in biopolymer-bound soil composites (BSC) while the material is wet (green). An impulse hammer taps the surface of a specimen to generate vibrations, and an accelerometer collects time-series response. The resulting time-series data are analysed with machine learning models, ExFeat and TsCheck.

ABSTRACT

Biopolymer-bound soil composites (BSC) are a novel class of cement-free building materials utilizing starch, protein, and lignin binders. While BSC are sustainable composite materials suitable for a wide range of construction applications, their manufacture is complicated, as quality issues (internal defects, improper mixing, improper compaction, etc.) may occur during manufacture. Even though conventional vision-based or acoustic-based quality control methods might be able to detect quality issues during the manufacture of BSC, they are unable to easily monitor the unique strength gain process of BSC that occurs when the wet material dries out. Conventional quality control methods are usually tailored to a single quality issue such as crack formation, requiring the use of multiple methods to completely verify material quality, which is inefficient. To address these issues we propose *BioSys*, a multi-functional quality control system to enable large-scale manufacture of BSC through non-destructive vibration-based testing. *BioSys* is multi-functional in the sense that it is used to: (1) identify internal defects (crack formation and improper mixing); (2) detect improper compaction; and (3) monitor desiccation. Unlike current methods, *BioSys* performs these tests in an efficient, non-intrusive manner, by generating signals from an impulse hammer tapping at different locations on a BSC specimen and measurement of response signals from an accelerometer. *BioSys* contains two different machine learning pipelines trained on the resulting time-series data with accuracy of up to 99% for defect detection and up to 100% for detecting improper compaction. *BioSys* reaches a MAPE of 5% for monitoring the strength gain of BSC during desiccation.

CCS CONCEPTS

• **Computing methodologies** → **Machine learning**; • **Hardware** → *Emerging architectures*; **Analysis and design of emerging devices and systems**.

BUILDSYS '24, November 7–8, 2024, Hangzhou, China

ACM ISBN 979-8-4007-0706-3/24/11
https://doi.org/10.1145/3671127.3698165

KEYWORDS

Quality control system, manufacturing, Defect detection, Sustainable materials, Biopolymer-bound soil composites

ACM Reference Format:

Barney H. Miao, Yiwen Dong, Andreas Theissler, Andrew Lesh, David J. Loftus, and Michael D. Lepech. 2024. BioSys: Efficient Quality Control System for Manufacturing of Sustainable Biopolymer Composites. In *The 11th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BUILDSYS '24), November 7–8, 2024, Hangzhou, China.* ACM, New York, NY, USA, 11 pages. https://doi.org/10.1145/3671127.3698165

1 INTRODUCTION

Sustainable materials are critical for the development of energy-efficient building systems that are designed to reduce carbon emissions. The development of biopolymer-bound soil composite (BSC) epitomizes a paradigm shift in sustainable construction practices by harnessing the desirable properties of biological binders to bind granular materials to form composites. BSC offers several benefits over conventional materials, such as: (1) having a negative life cycle carbon footprint, (2) providing new applications for underutilized and renewable biopolymers [24, 37], and (3) being circular materials with the potential to be recycled multiple times. BSC holds potential benefits over conventional building materials, but is not yet commercially available. Nevertheless, with additional development, BSC has shown significant potential to reduce energy consumption and carbon emissions associated with the built environment [36]. Therefore, to prepare for the introduction and possible large-scale manufacture of BSC in the construction world, techniques for quality control are needed.

Quality control is essential to ensure that materials used to construct the built environment have adequate strength, have consistent mechanical properties, are free of defects, and exhibit long term durability [21, 29, 32]. Overall, quality control helps optimize material use by reducing material waste, while also setting the initial health of a structure [43]. Improperly manufactured materials have been significant factors in structural failures, leading to a large loss of life. For example, the collapse of the Savar Building in Bangladesh and the collapse of the Surfside condominium in Florida resulted in 1,130 and 98 deaths, respectively. By performing quality control during material manufacture, we can help limit the occurrence of these catastrophic failures and help prolong the design life of our built environment.

Quality control of building materials has been performed by a wide assortment of tests ranging from physical — rebound hammer, penetration resistance, and pull-out [3, 14, 38, 39] — tests, ultrasonic pulse velocity [38], acoustic emission for defect detection [5], X-ray Micro-CT [42], digital image correlation [30], and vision-based methods [18, 41]. These tests are mostly non-destructive and, collectively, provide a diverse array of information regarding the health of building materials [23, 40]. Each individual test, however, probes just a single property of the material, necessitating several testing modalities to obtain a full picture of the material health. While the lack of a multi-functional quality control method is not an issue for lab development of BSC, it is an obstacle to large-scale manufacture of BSC, as the use of multiple testing methods for quality control is cumbersome and impractical. Furthermore, unlike concrete which gains strength through a chemical reaction called hydration [4], strength gain in BSC occurs through desiccation, where all the solvent that is used to initially dissolve biopolymer evaporates from the BSC [2, 37]. Laboratory methods for monitoring the strength gain of BSC involve tracking the loss of solvent during desiccation, i.e. monitoring the specimen's mass. However, tracking strength gain *via* mass-loss is only feasible for easily handled BSC products (e.g. bricks, tiles, etc.), but not for BSC products that are too large to be weighed or require manufacture *in situ* (pavement). Although conventional methods such as computer vision and acoustic emission are likely to be able to detect quality issues of BSC during initial manufacture, they are not able to monitor the strength gain of BSC. This is due to the uniqueness of the desiccation process and due to limitations of visual obstructions and frequency range constraints in the existing methods. This calls for an efficient, multi-functional, non-destructive quality control system for the large-scale manufacture of BSC.

To address these challenges, we introduce *BioSys*, an efficient, multi-functional quality control system. Unlike most existing NDT methods (primarily used for quality control of hard materials), *BioSys* is designed to perform quality control for composite materials while they are still in the wet state. Early stage detection of quality issues are important, as material in the wet state can be more easily salvaged, thus helping to reduce material waste from the manufacturing process. With *BioSys*, we apply a controlled impulse force (*via* an impulse hammer) onto the BSC surface. The resulting vibration waves propagate through the material and can be collected by a vibration sensor (e.g., an accelerometer) to infer the material properties of BSC and assess quality issues. *BioSys* is efficient in the following ways: (a) it uses off-the-shelf hardware, (b) the measurement system can be set up quickly, and (c) *BioSys* uses efficient machine learning methods, in both training and inference, so that rapid analysis can be performed. Furthermore, *BioSys* is multi-functional in the sense that it is used to (a) detect a variety of internal defects (e.g., regions with density anomalies and delamination/internal crack development), (b) ensure proper compaction of BSC, and (c) monitor the strength gain of BSC.

The operational procedure of *BioSys* is time efficient and convenient, as the accelerometer can be detached and reattached onto different BSC specimens quickly, similar to how a stethoscope is easily used to check the health of a patient. Additionally, *BioSys* is a location invariant system, as its ability to function is not significantly affected by the position of the accelerometer and different tapping locations on the BSC sample, as shown in our evaluation. To analyze the impulse and response data, we used two different time-series-based machine learning pipelines: (1) ExFeat: a simple high-level method using peak amplitudes and locations as features (8-dimensional feature space) for efficient assessment and (2) TsCheck: a more rigorous method that extracts additional features (20,000-dimensional feature space) for in-depth analysis. By incorporating two machine learning pipelines, *BioSys* can effectively perform quality control for the manufacture of BSC regardless of the number of features used to train the system.

In this paper, we make the following contributions:

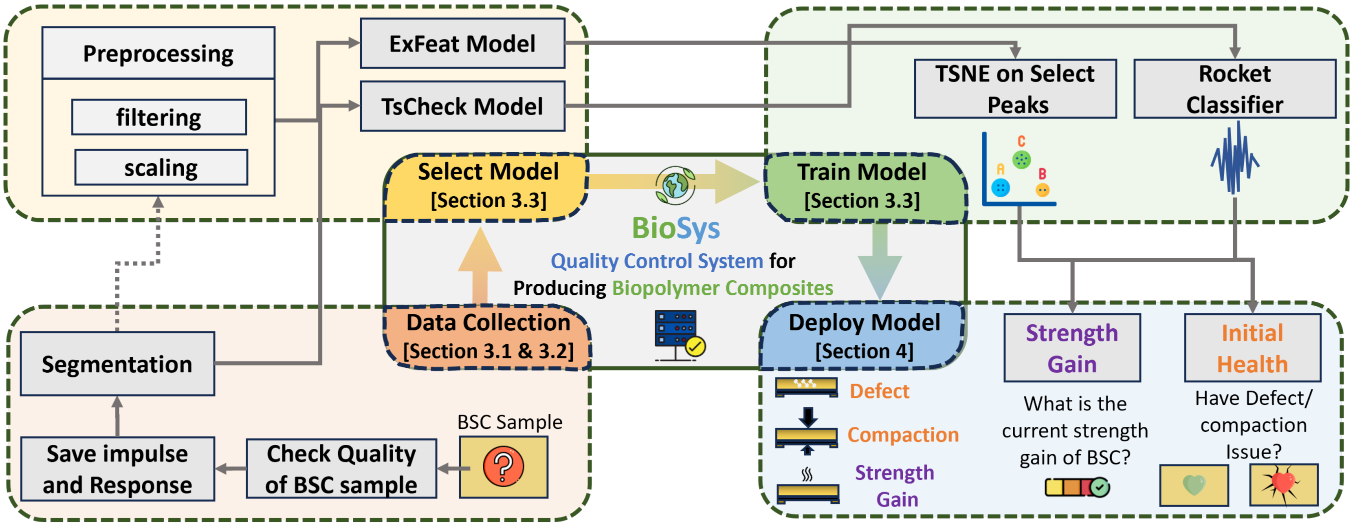

Figure 2: Overview of our *BioSys* system for quality control (initial health and strength gain) of biopolymer composite materials. *BioSys* uses data collected from vibration-based impulses on a BSC sample (Section 3.2). The impulse and response data is fed into ExFeat and TsCheck (Section 3.3), which are two models that allow *BioSys* to analyze the state of BSC at two different time-scales (Section 4). Depending on desired use, *BioSys* performs quality control of BSC (defect detection, mixing quality, proper compaction, and monitoring strength gain).

- Introduction of *BioSys*, the first quality control system to be used to evaluate a biocomposite material that undergoes solvent removal in order to gain strength.
- Deployment of *BioSys* in a study with 13 test specimens exploring different combinations of sensor and tap locations.
- Demonstration of how *BioSys* can reveal information regarding material cracking, mix inhomogeneity, compaction problems and strength gain.
- Our most impressive accomplishment was the detection of thin, low density defects as small as 100 microns modeled using plastic films.

2 BACKGROUND AND IMPLICATIONS OF BSC FOR ENERGY-EFFICIENT BUILDINGS

Biopolymer-bound soil composites (BSC) are a novel class of cement-free construction materials, that use biologically inspired binders sourced from the waste streams of major industries (paper and pulp production, bio-fuel industries, slaughterhouses, etc.) [22, 28, 37]. These biopolymers, when dissolved in solution, form a biopolymer matrix which binds together granular aggregates to form solid composite materials. Previous studies on BSC focused on the use of biopolymers such as various proteins (bovine serum albumin BSA, animal blood plasma AP920), polysaccharides (gellan gum & xanthan gum), and lignin, which is derived from plants [1, 6–8, 11, 17, 27, 35]. Unlike conventional building materials such as ordinary portland cement concrete (OPCC), BSC is a potentially carbon-negative material that promotes sustainability and energy efficiency of the built environment, as the use of carbon-rich biopolymers (BSA = $C_8H_{21}NOS_2 \approx 45\%$ carbon) in BSC are effectively a form of carbon sequestration [19, 24, 33]. Therefore, with the gradual large-scale adoption of BSC, we can make our buildings more energy-efficient at the construction stage. The compressive strength of BSC is comparable to conventional concrete, which makes BSC a viable cement-free alternative to concrete [37].

To manufacture laboratory-scale specimens of BSC, granular material and biopolymer are mixed for 5 minutes until thoroughly blended. Next, a solvent is added and further mixed for 5 more minutes to form the fresh mix [37]. The fresh mix is shaped and subjected to double-sided 5 MPa compactions using a custom molding apparatus and then desiccated in the open or through energy-enhanced methods (i.e. oven) until fully dried out. For large-scale manufacturing of BSC to be achieved, quality control systems will need to be developed.

3 OVERVIEW OF *BIOSYS* SYSTEM

BioSys is a multi-functional quality control system that aids the manufacture of novel sustainable biopolymer composite materials. *BioSys* detects the formation of internal cracks, determines mixing quality, checks uniform compaction, and monitors strength gain of BSC. Unlike quality control systems for conventional materials, which mostly focused on verifying the quality of the finished material [13, 31], the scope of *BioSys* covers initial material manufacture and strength gain stages. The *BioSys* system as shown in Figure 2 can be summarized in four main steps:

(1) **Data Collection and Preparation**: After manufacturing the BSC samples (Section 3.1), while they are still in a wet state, a controlled impulse force (e.g., impulse hammer tap) is exerted on the specimens to excite their internal structure. The responses are recorded by a vibration sensor (e.g., an accelerometer) for subsequent analysis (Section 4.1). As part

of the machine learning pipeline, data preparation steps are conducted.

(2) **Select Model**: We provide two machine learning pipelines, with the primary difference being in the number of features that are considered during training. The Exfeat model considers the most informative features extracted from the first four prominent peaks of the response data (8-dimensional feature space). The TsCheck model uses more sophisticated time-series models, which consider significantly more features (20,000-dimensional feature space) during training. By considering significantly more features, TsCheck typically has higher performance albeit with more computational requirements. For our final implementation, we used the ROCKET classifier for our TsCheck model based on the results shown later in Table 2.

(3) **Train Model**: For the ExFeat model, t-distributed stochastic neighbor embedding (t-SNE) was used to compress the peak amplitudes and locations for the response data. For the TsCheck models, ROCKET was used to perform feature extraction and quality control for BSC. The details about the model training and testing are introduced in Section 3.3.

(4) **Deploy Model**: With the trained models, *BioSys* is deployed during testing to perform quality control of BSC. For quality control, *BioSys* can identify internal defects (regions with density anomalies and delamination/crack formation). *BioSys* can determine if BSC samples are uniformly compacted during manufacture and can track the strength gain of BSC.

3.1 Sample Manufacturing

A series of custom molds were fabricated through 3D printing of polylactic acid (PLA), to help shape the specimens during manufacture and to simulate two separate low-density defects. One of the 3D printed objects was a hollow sphere, and the other was a hollow cylinder, 1 mm wall thickness for each (Sphere_V and Cylinder_V, from Figure 4). We also created high density defects by using a steel sphere and steel cylinder. The dimensions of the two low density defects were the same as the high density defects. For the final low-density defect, Circle_V (Figure 4), a plastic petri dish with lid was used to create a small cylindrical void in the middle of the sample. The density contrast between the high-density defect (7.85 g/cm^3) and BSC (1.40 g/cm^3) is higher than what might be anticipated, such as from mixing inhomogeneity, it is still a valuable test to gauge the performance of *BioSys*.

3.2 *BioSys* Data Collection and Preparation

To collect the data used to train *BioSys*, a controlled impulse input (from an impulse hammer) is used to tap various locations on the surface of a BSC specimen. The resulting vibration is recorded by a vibration sensor (e.g., an accelerometer) affixed to the BSC specimen. The location of the vibration sensor on the BSC specimen is varied, to determine if the position of the sensor or location of tapping affects the performance of *BioSys*. To isolate the specimen from background noises, we place the BSC specimens on a dynamically isolated platform, which helps ensure that the measured response is directly related to the BSC instead of the surface it rests on. The analog signal is converted to a digital signal through a commercial A-D converter and recorded.

Formally, the measurements obtained with *BioSys* (see Figure 6) correspond to a multivariate time series x_T. See Definition (3.1):

Definition 3.1. A **time series** $x_T : \overrightarrow{T} \mapsto \overrightarrow{X}$ is a finite sequence of N data points ordered by time. For each time point t_i in $\overrightarrow{T} = \{t_1, ..., t_N\}$, there is one data point x_i in $\overrightarrow{X} = \{x_1, ..., x_N\}$ in the case of **univariate time series**, and $d > 1$ data points in the case of **multivariate time series**. The time points t_i are used to index the time series, where $x_{T_{t_i \ldots t_{i+k}}}$ denotes a **subsequence** s_i.

For *BioSys*, the multivariate time series x_T consists of two univariate time series: (1) The **hammer time series** x_{T_h} which holds the pressure measurements obtained from the impulse hammer when tapping the hammer on the specimen. (2) The **response time series** x_{T_r} which holds the vibrations associated with tapping of the hammer measured by the accelerometer.

The machine learning pipeline encompasses two different approaches: (1) manual feature extraction which we denote as ExFeat and (2) application of time-series machine learning models which we denote as TsCheck. The time series are first segmented, followed by an optional preprocessing step that is specific to ExFeat and TsCheck:

Segmentation into subsequences. From the time series x_T, subsequences s_i are extracted such that each subsequence contains one tapping of the hammer and its corresponding response. This is done by identifying and noting the times of each individual peak in the hammer time series data. Since each identified peak is associated with an individual hammer tap, we can use the noted times to locate the peaks in the response time series data. Once each peak in the response data has been located, a window is defined, from which segmentation of the response data is conducted to obtain subsequences of the response times series associated with each measured hammer impulse (please see Figure 6 and Section 4.6.2). Note that s_i is in turn a multivariate time series. Segmentation is achieved by

(1) find the time points t_{peak} of the peaks in the hammer time series
(2) for each peak, obtain a subsequence s_i determined by T_{pre} and T_{post} which specify the number of time points prior and after the detected peak t_{peak} respectively. Hence, $s_i = x_{T_{(t_{peak}-T_{pre}) \ldots (t_{peak}+T_{post})}}$.

Preprocessing by filtering [ExFeat Model only]. Since the magnitude of the tapping may vary, for the ExFeat model the subsequences s_i are filtered such that the remaining subsequences are within 0.9 and 1.1 of the mean hammer impulse measured across all the samples. Due to the hammer impulses being manually applied to the BSC specimens, the impulse imparted on the specimens will vary. Thus, to account for differences related to varying hammer impulse, responses were filtered such that the response data used for training and testing of the ExFeat model will have similar hammer impulse force. This is done by determining the maximum value (peak) of each hammer subsequence, and selecting the subsequences that have a hammer peak that is within 0.9 and 1.1 times from the arithmetic mean value of all peaks. A narrower filtering window typically results in a more accurate model, as the selected response subsequences have increasingly similar hammer impulses. However, this benefit begins to fall off when too many subsequences are filtered out. Therefore, the filter range (filters out approximately 80% of the segmented response subsequences) was determined based on balancing this trade-off between improved accuracy and number of subsequences available for training the model. The filtering procedure is achieved as follows

(1) for each subsequence s_i determine the maximum value of the hammer subsequence $x_{T_{h_{peak}}}$
(2) filter subsequence by selecting those that lie in the range $[0.9\cdot\overline{x_{T_{h_{peak}}}}, 1.1\cdot\overline{x_{T_{h_{peak}}}}]$, where $\overline{x_{T_{h_{peak}}}}$ denotes the arithmetic mean value of all peaks

Preprocessing by scaling [TsCheck Model only]. Similar to the ExFeat model, variations in the hammer impulse can be accounted for by scaling the response of each subsequence s_i with respect to the hammer impulse. This is achieved as follows

(1) for each subsequence s_i determine the maximum value of the hammer subsequence $x_{T_{h_{peak}}}$
(2) scale the response subsequence x_{T_r} as follows $x_{T_r} := \frac{x_{T_r}}{x_{T_{h_{peak}}}}$

3.3 Model Selection and Training in *BioSys*

Upon collecting the data shown in Table 1, we can now develop ExFeat models (considers an 8-dimensional feature space) and TsCheck models (considers a 20,000-dimensional feature space) for defect detection, monitoring strength gain, and ensuring proper compaction. By adding additional analysis options to *BioSys* we can increase the flexibility and versatility of the system, as it can deliver effective results in two realistic time frames (albeit with a trade-off between accuracy and computational costs), and demonstrate that *BioSys* can effectively perform quality control of BSC manufacture through two different model approaches.

3.3.1 ExFeat model development. We develop the ExFeat model to perform quick quality control by extracting the first four peak amplitudes and locations from the response time series. Initial visual inspection of the data shows no obvious difference between a normal BSC sample and one with an internal defect.

However, if we focused on individual taps (Figure 3), we can discern more noticeable differences in the time-series response between normal BSC and defect BSC specimens. These features are further compressed into two dimensions through a Euclidean t-SNE operation (Figure 7), to get the two features (1st and 2nd t-SNE dimensions) that train the ExFeat model. Machine learning classifiers for tabular data (i.e. feature spaces) are evaluated: e.g. support vector machines with different kernels, nearest neighbors, decision trees, and neural networks.

3.3.2 TsCheck model development. While the ExFeat model described above is a fast and simple approach, it has two inherent limitations: (1) it requires manually specifying the features to be extracted, so-called feature engineering, and (2) the temporal aspect of the data is only considered to a limited extent. Hence, as

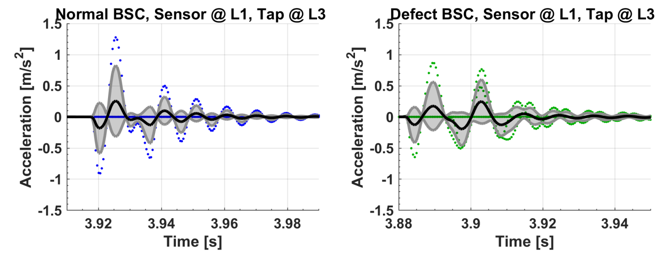

Figure 3: Sample response time series x_{T_r} for normal BSC samples and those with defects. The signals from the samples with modeled defects were easily distinguishable from the normal samples. The individual scatter points represent the times series points for all five 10-second time interval tests at the given interval shown, while the black line and grey envelope represent the average response and standard deviation respectively, for all five 10 second time interval tests.

an alternative, we tested a wide range of machine learning models (classifiers) that work directly on the time series and do not require manual feature engineering. We evaluated models with different fundamental concepts, all of which were used in their implementation in *aeon, a Python toolkit* [25]:

- distance-based methods that classify based on the distances between the considered time series segments: 1-Nearest Neighbors using Euclidean distance as well as dynamic-time-warping distance
- feature-based methods that extract a number of pre-defined features and classify on the feature space: TSFresh [9] and Catch22 [16]
- ensemble-based methods which combine different base classifiers to make a decision: Random Interval Spectral Ensemble (RISE) [12] and HIVECOTEV2 [26] (with two minutes training time budget)
- a kernel-based classifier exploiting the ideas of kernels (windows) moved over the time series, similar to a 1-dimensional convolutional neural network. We used the ROCKET classifier [10] which, instead of learning the kernel parameters (weights, bias, dilation and padding) from the data, uses a high number of kernels with randomly sampled parameters. A linear classifier is then trained on the ROCKET-induced feature space. Due to not learning the kernel parameters and due to using a simple classifier, this method is computationally very efficient.

4 EVALUATION OF *BIOSYS*

For this study, 13 specimens (120 mm ±2 *mm* long x 90 mm ±1 *mm* wide x 25 mm ±1 *mm* depth) of protein-based BSC (AP920 as the biopolymer) were manufactured according to a single mix design (300-gram sand, 50-gram water, 50-gram AP920). Within these samples, two different manufacturing styles — without mold (group 1) and with mold (group 2) — were tested to see if manufacturing style had an impact on the performance of *BioSys*. Data was collected at different combinations of sensor location and tap location in order to determine if defect detection of *BioSys* is location dependent. Figure 4 shows the types of specimens fabricated in this study. Table 1 gives an overview of the recorded data.

Two normal specimens were fabricated for each manufacturing style. "Normal" in this study refers to BSC samples with no known defects. Eight different samples with various modeled defects were fabricated as shown in Figure 4. Regions with higher density can occur within BSC in the form of mixing inhomogeneities or random placement of unusually large aggregate pieces or contaminants. To simulate these high-density regions, we embedded a steel ball (d = 19 mm) and steel cylinder (d = 12.5 mm and h = 75 mm) into the BSC specimen. Similarly, to detect regions with low-density anomalies (regions with large voids or lack of binder), hollow 3D printed spheres/cylinders (1 mm thick walls) or plastic petri dishes with lids (1 mm thick wall, d = 40 mm and h = 12 mm) were used. Finally, to simulate regions with internal cracks or delamination, thin plastic films were placed in the middle of the specimen, with different orientations and sizes of defects being explored (Diag_C: h = 10 mm and w = 20 mm, Ssquare_C: square 30 mm x 30 mm, and Bsquare_C: square 70 mm x 70 mm). The thickness of the plastic films was around 100 microns, which is smaller than the maximum allowed flaw size of 150 microns for concrete under different exposure conditions (ACI committee 224, AASHTO LRFD, BS-8110, and Eurocode 2) [34]. Hence, if *BioSys* can detect the presence of a piece of plastic film, it should be able to detect cracks in BSC. For detecting proper compaction and strength gain, three additional normal specimens were manufactured, with two of the normal samples being used for evaluation of single-sided compaction versus double-sided compaction, and the third normal sample being used for the monitoring of strength gain of the sample over the course of desiccation.

4.1 Experimental Setup and Data Collection

Figure 5 shows the experimental setup for collecting data from BSC samples in this study. A small rectangular platform was dynamically isolated from the ground, by placing four foam dampers on the corners of the platform. Wax glue was used to affix the BSC sample to the platform, and later to secure an accelerometer (sampling frequency = 2048 Hz) to measure the z-axis response at location 1 (bottom left corner and tap at locations 2, 3, and 5) and location 5 (middle of sample and tap at locations 1, 2, 3, and 4), as shown earlier in Figure 4.

The accelerometer used in the experiment is the PCB Piezotronics tri-axial accelerometer (Model No.354C03). The sensor was connected with the NI-DAQ to convert the analog signal to a digital time series. Each impulse force was applied to the specimens by a PCB impact hammer (Model No.086D20) for ease of use in laboratory settings. For these experiments, impulse force using the hammer was applied manually. Two different operators were involved in using the impulse hammer to tap the BSC specimens. There was no significant difference in the results obtained from one operator compared to the other operator. At each combination of sensor and tap location, data was collected for five 10-second time intervals, in which the impulse hammer is continuously tapped on the BSC for around 10 - 20 times per interval. Table 1 shows the

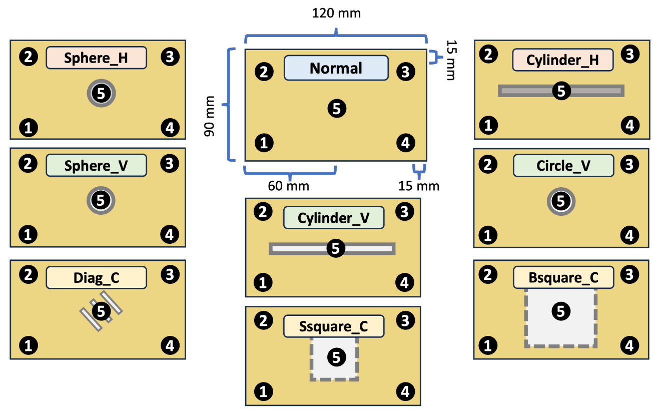

Figure 4: Top down view of test specimens. Specimen Sphere_H contains a solid sphere (i.e, steel ball). Specimen Sphere_V contains a hollow plastic sphere. Specimen Diag_C (to model cracks) contains three plastic films arranged as shown. Specimen Cylinder_V contains a hollow plastic cylinder. Specimen Ssquare_C contains a single piece of plastic film. Specimen Cylinder_H contains a solid steel rod. Specimen Circle_V contains a plastic petri dish with lid. Specimen Bsquare_C is similar to specimen Ssquare_C but the plastic film is larger. All modeled defects are located approximately in the middle of the specimen (along z-axis).

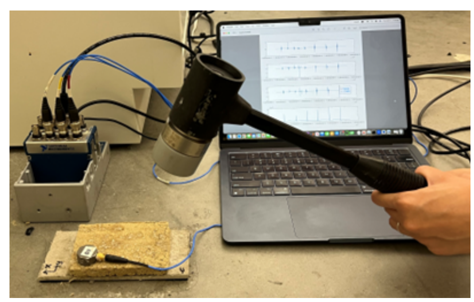

Figure 5: *BioSys* data collection involves the use of an impulse hammer to tap the BSC specimen. The vibration response is recorded by an accelerometer.

total number of taps imparted on each type of BSC sample. Figure 6 shows an example of the time-series data collected from *BioSys*.

For defect detection, the accelerometer was attached to the platform at two locations: (1) location 1 (bottom left corner) and (2) location 5 (center). For group 1 defects, we collected data at both of these locations, to see if the accelerometer had to be over the defect for *BioSys* to detect it. However, upon analyzing the results, we found that *BioSys* had better performance when the accelerometer was placed at location 1 instead of location 5. Hence, for the remaining data collection for group 2 defects, we only collected data from location 1. The impulse hammer was used to tap locations 2, 3, and 5 when the accelerometer was placed at location 1, while locations 1, 2, 3, and 4 were tapped when the accelerometer was at location 5.

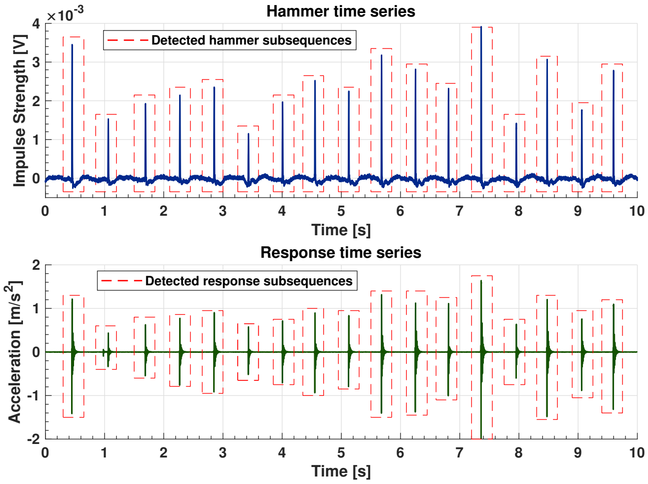

Figure 6: For these experiments impulse force using the hammer was applied manually and the strength of the impulse was measured as a voltage. The red dashed boxes schematically illustrate the extraction of subsequences from the time series data (width of subsequences is determined by the parameters T_{pre} and T_{post} – used subsequence width is much smaller than illustrated in this example).

Table 1: Total individual sequences measured per sample

BSC Type	Fabrication	Total number of taps ([S1],[S5])*
Normal	No Mold [1]	244 ([S1:76/84/84])
Normal	With Mold [2]	577 ([S1:93/76/88],[S5:79/83/75/83])
Sphere_H	No Mold [1]	568 ([S1:85/81/83],[S5:91/84/75/69])
Cylinder_H	No Mold [1]	602 ([S1:90/90/90],[S5:81/86/79/86])
Sphere_V	With Mold [2]	223 ([S1:73/74/76])
Cylinder_V	With Mold [2]	233 ([S1:79/78/76])
Circle_V	No Mold [1]	610 ([S1:86/89/87],[S5:90/89/81/88])
Diag_C	With Mold [2]	203 ([S1:63/71/69])
Ssquare_C	With Mold [2]	241 ([S1:79/80/82])
Bsquare_C	With Mold [2]	217 ([S1:74/74/69])
Normal	Strength Gain	2hr-7hr [66/59/83/70/70/70] **

* [Sen @ 1: tap @ 2 / tap @ 3 / tap @ 5], [Sen @ 5: tap @ 1 / tap @ 2 / tap @ 3 / tap @ 4] as shown in Figure 4
** The number of taps on the same sample when it is 29 %, 39 %, 51 %, 63 %, 74 %, and 80 % desiccated or developed.

In order to evaluate BSC compaction, which is a necessary step in the manufacture of specimens, an accelerometer was placed at location 1, with tapping being conducted at location 5 of the specimen. Upon finishing five 10-second time intervals of tapping, the sample was flipped, with the procedure repeated to collect data for the other side.

For measuring strength gain, a normal sample (no defect) was attached to the dynamically isolated platform. An accelerometer was fixed at location 1, with tapping occurring at location 5 of the specimen for five 10-second time intervals. This procedure was repeated for a total of five 10-second time intervals for each of the six desiccation levels (see Table 1), with the mass of the BSC also being recorded during each interval, to help determine the degree of desiccation of the BSC sample.

4.2 Data Analysis

We evaluated *BioSys* for ExFeat and TsCheck models on 13 BSC samples (10 samples for defect detection by combining fabrications group 1 and group 2, 2 samples for proper compaction, and 1 sample for monitoring strength gain). In this section we summarize the results of *BioSys* in detecting defects (cracks and proper mixing), detecting proper compaction, and monitoring strength gain. In our implementation of *BioSys* the train/test split for both ExFeat and TsCheck was 80 % / 20 %.

4.3 Results: Defect Detection

Upon collecting data on the two groups of BSC samples (Table 1) according to Section 4.1, *BioSys* was able to differentiate between normal BSC and BSC with various defects fabricated with a mold (confined applications: such as tiles, bricks, etc.) or without a mold (unconfined applications: BSC in additive construction).

To obtain quick results we use our ExFeat model. After evaluating several standard classifiers, ExFeat uses a support vector machine (SVM) with Gaussian kernel on the two t-SNE dimensions (see Figure 7) calculated from the amplitudes and locations of the first four peaks of individual subsequences. If we used ExFeat to determine "defect vs. no defect" (i.e., we don't specify the type of defect), the ExFeat model obtains an accuracy of 66% without filtering. We refer to this as "2-class" classification. If we used *BioSys* to determine individual defects ("9-class" classification), the ExFeat model obtains an accuracy of 47% without filtering. ExFeat, however, achieves an accuracy of 80% for 2-class and 70% for 9-class classification after filtering, as described in Section 3.2.

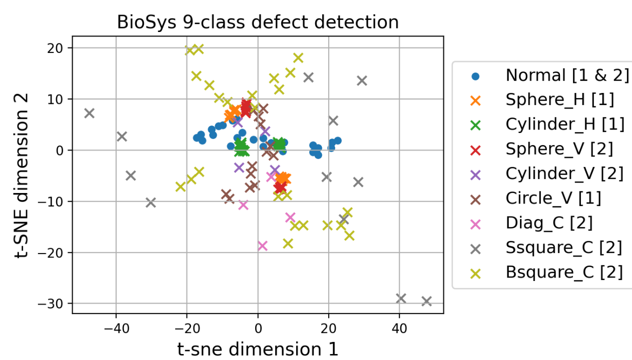

Figure 7: t-SNE of filtered sequence peak locations and amplitudes for ExFeat model. The blue circles in the middle of the plot represent normal BSC samples, while the multi-colored x's represent BSC with various defects.

To enhance the performance of *BioSys*, we use our TsCheck model. Based on the classifiers' accuracies (see Table 2) and its computational efficiency, the final version of the TsCheck model uses ROCKET (with the default number of 10,000 kernels yielding a 20,000-dimensional feature space). In comparison to the ExFeat

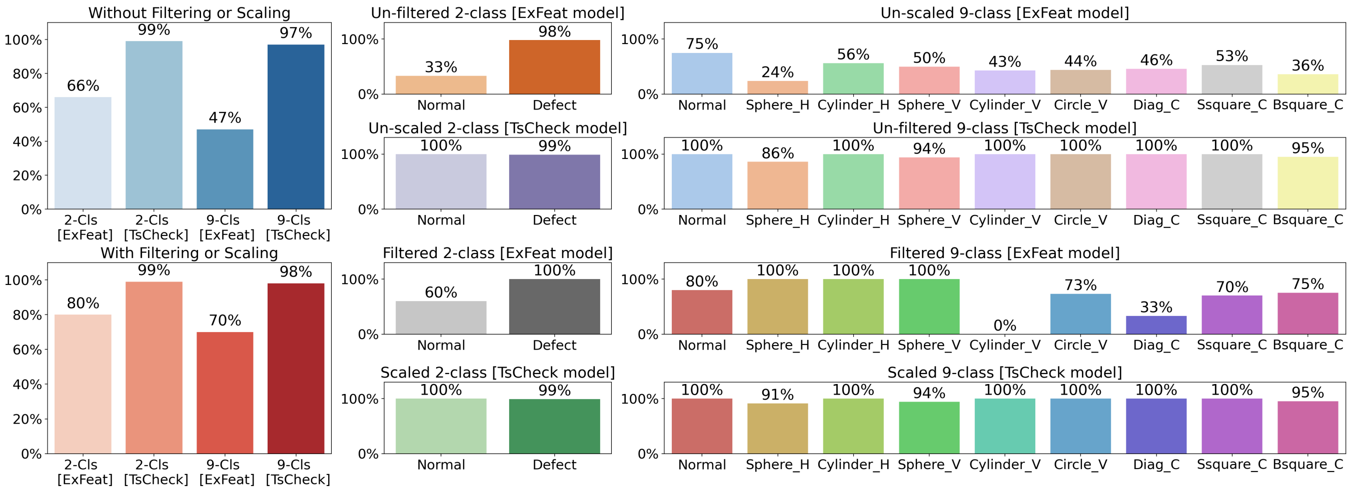

Figure 8: Accuracies of ExFeat model and TsCheck (using ROCKET) model on defect detection within BSC (combining fabrications group 1 and group 2). As shown, the TsCheck model has superior performance compared to the ExFeat model (2-class and 9-class). By filtering or scaling the response subsequence by the force of the impulse hammer we could further improve the accuracy.

Table 2: Test Results of *BioSys* comparing ExFeat Model [t-SNE] with other contenders for the TsCheck Model [ROCKET] for detecting defects during the manufacture of BSC. (Based on the combined data from both fabrications group 1 and group 2; averaged accuracies based on test data from Sen @ 1 : tap @ 3; class ranges (R) are the accuracies of the best and worst class).

Defect detection (data not preprocessed)				
Method	2-Class	2-Class Range	9-Class	9-Class Range
ExFeat Mdl	**66 %**	**33 % - 98 %**	**47 %**	**24 % - 75 %**
KNN (1-NN)	91 %	83 % - 99 %	78 %	50 % - 100 %
TSFresh	91 %	83 % - 99 %	86 %	75 % - 100 %
Catch22	85 %	73 % - 96 %	72 %	45 % - 100 %
RISE	90 %	78 % - 100 %	89 %	72 % - 100 %
HIVECOTEV2	90 %	80 % - 99 %	96 %	89 % - 100 %
TsCheck Mdl	**99 %**	**99 % - 100 %**	**97 %**	**86 % - 100 %**

ExFeat marco-averaged F1 score: 2-class = 0.72 and 9-class = 0.52
TsCheck marco-averaged F1 score: 2-class = 0.99 and 9-class = 0.97

Defect detection (data preprocessed)				
Method	2-Class	2-Class R	9-Class	9-Class R
ExFeat Mdl	**80 %**	**60 % - 100 %**	**70 %**	**0 % - 100 %**
KNN (1-NN)	91 %	83 % - 98 %	76 %	54 % - 100 %
TSFresh	90 %	83 % - 97 %	87 %	72 % - 100 %
Catch22	87 %	77 % - 97 %	69 %	31 % - 90 %
RISE	82 %	63 % - 100 %	86 %	72 % - 97 %
HIVECOTEV2	98 %	97 % - 99 %	98 %	91 % - 100 %
TsCheck Mdl	**99 %**	**99 % - 100 %**	**98 %**	**91 % - 100 %**

ExFeat marco-averaged F1 score: 2-class = 0.92 and 9-class = 0.80
TsCheck marco-averaged F1 score: 2-class = 0.99 and 9-class = 0.98

model, which on a standard PC takes less than a minute to train (around 30-40 seconds). While the final TsCheck model has fast execution times, finding the best model among the tested model candidates was much more computationally expensive. Figure 8 shows the performance of the TsCheck model achieving 99 % for 2-class and 97 % for 9-class classification on the unscaled data.

During data collection, an impulse hammer was used to tap the BSC sample, with the response being collected by an accelerometer attached to the surface of the specimen. However, as the force imparted on the specimen varies between each impact, the measured vibration will also vary. This effect is especially important to consider in the ExFeat model, where the variation in peak amplitudes in the response data directly affects half of the selected features. Hence, if we train the ExFeat and TsCheck models on response data that has been scaled according to the impulse force, we will be able to improve the performance of *BioSys* further. Figure 8 shows the beneficial effect of scaling (normalization) of the measured response from the accelerometer by the impulse hammer force and the superior performance of the TsCheck model over the ExFeat model. Using a 9-class approach across the various types of defects that we modeled in the material, we achieved defect detection rates ranging from 91% to 100%. The TsCheck model shows dramatically better accuracy than the ExFeat model — 97% accuracy versus 47% — without filtering or scaling. Even with filtering, the best performance of the ExFeat model is 70%. With scaling the TsCheck model shows further improvement to 98%. All of these comparisons are for 9-class classification. The advantage of preprocessing the data is also shown in the increase in F1 scores, with both models having well-balanced performance when variations in the hammer impulse are accounted for. See Figure 8 to compare the performance of ExFeat and TsCheck for 2-class and 9-class classification.

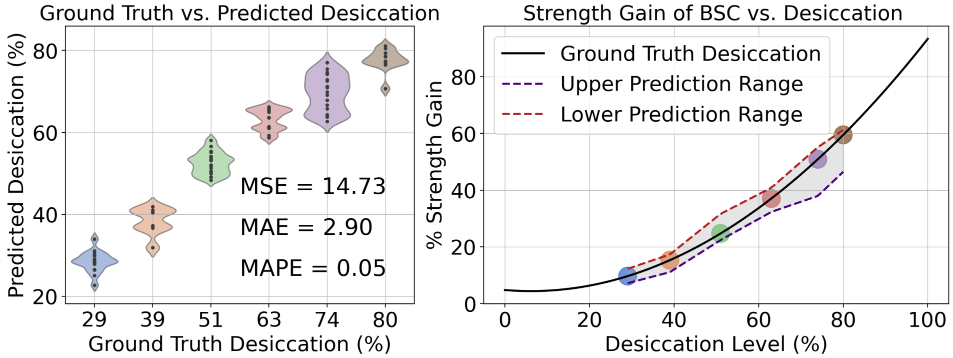

Figure 9: Predicted desiccation vs. ground truth desiccation of BSC sample. From the desiccation level we can infer the level of strength gain, from where we can plot the upper and lower predicted strength gain curves.

4.4 Results: Compaction Quality Control

To determine the effectiveness of *BioSys* towards detecting proper compaction, we can compare the transfer functions of the two sides of the BSC. Intuitively, a BSC sample that has been subjected to single-sided compaction vs. double-sided compaction will not be uniform based on TsCheck model classification (again using ROCKET as a classifier). Hence, the measured response/ transfer function on each side of the BSC will not be the same. Using our approach, *BioSys* can determine if BSC materials have been properly compacted during manufacture, with almost 100 % test accuracy. Results not shown.

4.5 Results: Strength Gain Classification

Monitoring strength gain is an essential step in the manufacture of BSC. In this study, we evaluated specimens of BSC over the course of complete desiccation, at various intervals. Even using the "quick and dirty" ExFeat model, *BioSys* achieved an accuracy of 77 % in evaluating degree of desiccation. Using the TsCheck model, *BioSys* achieved an accuracy of 100 %. Figure 9 shows the regression results (using ROCKET as a regression model) obtained from the TsCheck model and converting the desiccation level to strength gain according to the relationships found earlier by Biggerstaff et al [2]. From the Figure, we can see that the desiccation level and strength gain curve are close to their expected ground truth values, where the regression model yielded a MAPE of 5% for the prediction of the desiccation based on the time series data.

4.6 Additional Considerations

During the development of *BioSys*, additional considerations are made for challenges that may affect the effectiveness of the model. These challenges include variations between samples, defining the subsequence size of individual responses, and the effect of data collection locations.

4.6.1 Variation between Samples. Although we used only 13 different BSC samples in our investigation, previous studies using this same material indicate that the variation from sample to sample is quite small [37]. Thus, we are confident that the data we have presented on the performance of *BioSys* detected real differences amongst the various specimens with modeled defects. This assertion is supported by the fact that our models could identify some differences between a normal BSC specimen made without a mold versus a normal BSC specimen made with a mold, but not to the extent of BSC with modeled defects. To determine if *BioSys* is distinguishing between different defects, rather than merely distinguishing between different bricks, we can train the TsCheck models with data from specimens made without a mold (normal specimen made without mold and defect specimens made without a mold) and make inferences based on the data from specimens made with a mold (normal specimen made with a mold and defect specimens made with a mold). Upon performing this test (using the TsCheck model with scaled data), *BioSys* was still able to perform 2-class defect detection, albeit with a noticeable reduction in accuracy ROCKET gives 72-79 % instead of 99 %). This is not surprising, as we are using a different fabrication method (with or without mold) between the two groups of samples, which in itself will naturally lead to some differences between the two groups (i.e. normal BSC specimens made without a mold should be different than normal BSC specimens made with a mold). Additionally, the defects learnt from primarily high-density defects are significantly different than the defects learned (trained) from low-density defects and internal cracks. This analysis gives us confidence that *BioSys*, can distinguish between specimens with different modeled defects and normal specimens. *BioSys* is not detecting trivial differences between specimens.

4.6.2 Defining the subsequence of individual responses. An important consideration to make during the development of *BioSys* is in defining the subsequence length of individual responses. During each 10-second interval of data collection, the impulse hammer was tapped approximately 10 - 20 times, resulting in a recorded response with the same number of individually distinguishable responses. The challenge here would be in the extraction of each response, as improper extraction may result in negatively impacting the performance of *BioSys*. Defining the start (left-bound of the subsequence, i.e. T_{pre}) of an individual response was not challenging – the start of an impulse is easily captured as the amplitude of the response was at its highest point early on.

However, defining the end (right-bound of the subsequence T_{post}) was challenging, as it was difficult to distinguish the end of a sequence, i.e. when the response signal has dissipated. For all of the implementations shown earlier for defect detection and later for strength gain classification and compaction quality control, we cut off the response such that the total length of an individual response profile was 0.08 seconds. This cut-off location typically occurs around the 8th min-max peak of a given response, where the amplitude of the peaks on the right-bound of the subsequence has now dissipated to 5-10 % of the max amplitudes (see Figure 3). Although this defined cut-off will not impact the results of the ExFeat model (which uses the amplitudes and locations of only the first four min-max peaks), it will impact the performance of the TsCheck model. To investigate any possible positive or negative impacts of subsequence segmentation, we expanded the subsequence length to 0.16 and 0.32 seconds, from 0.08 seconds.

The TsCheck model with expanded subsequence size did not affect the results of HIVECOTEV2 and ROCKET (final choice for TsCheck model) classifiers but slightly improved the 2-class accuracies of the other time series models.

4.6.3 Impact of sensor and tap location on performance of BioSys. The 2-class accuracy of defect detection of unmolded specimens with the TsCheck model dropped from 99 % to 95 % when training on all data from sensor 5 instead of sensor 1. Therefore, as *BioSys* can detect defects regardless of which set of sensors and tap location is used to train the model, we conclude that the system is location invariant. If TsCheck was trained with data from sensor @ location 1 and tested with data from sensor @ location 5, we find that model accuracy drops to random guessing. However, if data from sensor @ location 1 and sensor @ location 5 were both used for training, the TsCheck model would be able to detect defects with an accuracy of up to 97 %. Therefore, *BioSys* can detect defects with high accuracy for tap locations and sensors it has seen before and fails to detect defects on data from new sensors or tap locations.

5 DISCUSSION

BioSys addresses a challenge that is inherent in the manufacture of BSC products — namely, the need to detect manufacturing defects in the material while it is in the green (wet) state. The advantage of detecting defects in this early state of manufacture is that the wet material can be returned to a point upstream in the manufacturing process so as to minimize waste. While there exists a wide range of non-destructive quality control methods, most are tailored for quality control of hard materials. We have designed *BioSys* to perform quality control at the initial state of manufacture of BSC. Thus, the challenge is to detect flaws in a soft material, which is rarely the arena in which non-destructive testing is applied. An example of a non-destructive technique for evaluation of soft materials is the use of guided elastic wave propagation using optical coherence elastography for characterizing biofilms [20]. Another example is the use of microwave holographic subsurface technology to detect void spaces in polyurethane foams [15]. Our study introduces a novel vibration-based sensing system that is designed specifically for evaluation of BSC in its wet state, thus filling the current gap left by conventional methods.

The investigation we present in this paper is our first foray into developing a novel technology for defect detection in wet BSC. To carry out this study, we selected a set of "defect BSC specimen" based on embedding conveniently available objects into the green BSC. Some of these objects are fairly large and may not adequately represent real world defects that may occur in manufactured BSC products. We recognize that more realistic defect models, of a variety of sizes, may be helpful to properly evaluate the performance of *BioSys*. In addition, larger numbers of specimens would be helpful to provide greater confidence in the performance of this novel system.

While *BioSys* can detect the existence of a defect in BSC, it cannot identify where the defect is located within the BSC specimen. In a manufacturing setting, it may not be necessary to identify the location of a defect if return of the material to an upstream point in manufacturing is the sole goal. Under scenarios where locating the defect is important, this might be accomplished by collecting additional data from different sensor and tap locations. This may be worth addressing for certain types of BSC products that are large and expensive, where salvage of a portion of the product, rather than discarding it completely, may make sense. Further collection of data will help test the limits of vibration-based sensing. For instance, one future experiment may involve increasing the separation distance between sensor or tap locations. This may be important for BSC objects that are larger or have a layered structure. It is even conceivable that vibration-based testing may be used for quality control of BSC that is manufactured using a 3D printing process.

6 CONCLUSION

In this paper, we introduce *BioSys*, which is an efficient quality control system to help aid the manufacture of sustainable biopolymer composites for the built environment. *BioSys* can accurately determine the initial health of BSC, during the green (wet) state. The ExFeat model, the streamlined version of *BioSys*, allows users to infer the state of BSC with results that are up to 70 % accurate in 9-class defect detection and 77 % accurate in strength gain characterization. The TsCheck model, the more complete version of *BioSys*, achieves up to 98 % accuracy in 9-class defect detection and up to 100 % accuracy in strength gain characterization. Additionally, the F1 scores of the two models indicate that *BioSys* has a well-balanced performance, especially after preprocessing the data. Therefore, based on the results of this study, *BioSys* shows excellent promise in aiding the introduction and possible large-scale manufacture of BSC, a material that will help to reduce the carbon footprint associated with constructing the built environment.

While the development of *BioSys* represents a significant step toward introducing an efficient multi-functional quality control system for the manufacture of BSC, there remains future work and avenues for applications that will help enhance the performance and impact of *BioSys*. One way of implementing *BioSys* in large-scale manufacture of BSC would be to use *BioSys* to evaluate freshly molded, still green products (e.g. blocks). The discovery of defects in products, before desiccation takes place, would allow identification of substandard products that would perform poorly in their design applications. In addition, the discovery of defects prior to desiccation would allow these products to be recycled at a point during the manufacturing process where recycling can be accomplished easily, so that the raw materials are not wasted. *BioSys* could also be a valuable tool for optimizing manufacturing processes. Close monitoring of quality during the manufacturing process could extend the design life of BSC products, further reducing life cycle impacts of this novel construction material.

7 ACKNOWLEDGMENTS

This work was supported by Shell International Exploration and Production inc. through its membership in the Stanford Strategic Energy Alliance. The sponsorship is gratefully acknowledged. Facility support for this work has been provided by the John A. Blume Earthquake Engineering Center at Stanford University.

REFERENCES

[1] Mohamed K Ayeldeen, Abdelazim M Negm, and Mostafa A El Sawwaf. 2016. Evaluating the physical characteristics of biopolymer/soil mixtures. *Arabian Journal of Geosciences* 9 (2016), 1–13.

[2] Adrian Biggerstaff, Michael Lepech, and David Loftus. 2022. Determining the structuration of biopolymer-bound soil composite. *Materials and Structures* 55, 7 (2022), 190.

[3] Denys Breysse and Juan Luis Martínez-Fernández. 2014. Assessing concrete strength with rebound hammer: review of key issues and ideas for more reliable conclusions. *Materials and structures* 47 (2014), 1589–1604.

[4] Jeffrey W Bullard, Hamlin M Jennings, Richard A Livingston, Andre Nonat, George W Scherer, Jeffrey S Schweitzer, Karen L Scrivener, and Jeffrey J Thomas. 2011. Mechanisms of cement hydration. *Cement and concrete research* 41, 12 (2011), 1208–1223.

[5] Alberto Carpinteri, Giuseppe Lacidogna, Gianni Niccolini, and Stefano Puzzi. 2008. Critical defect size distributions in concrete structures detected by the acoustic emission technique. *Meccanica* 43 (2008), 349–363.

[6] Ilhan Chang and Gye-Chun Cho. 2014. Geotechnical behavior of a beta-1, 3/1, 6-glucan biopolymer-treated residual soil. *Geomech. Eng* 7, 6 (2014), 633–647.

[7] Ilhan Chang, Jooyoung Im, Awlia Kharis Prasidhi, and Gye-Chun Cho. 2015. Effects of Xanthan gum biopolymer on soil strengthening. *Construction and Building Materials* 74 (2015), 65–72.

[8] Ilhan Chang, Minkyung Jeon, and Gye-Chun Cho. 2015. Application of microbial biopolymers as an alternative construction binder for earth buildings in underdeveloped countries. *International journal of polymer science* 2015 (2015), 1–9.

[9] Maximilian Christ, Nils Braun, Julius Neuffer, and Andreas W Kempa-Liehr. 2018. Time series feature extraction on basis of scalable hypothesis tests (tsfresh–a python package). *Neurocomputing* 307 (2018), 72–77.

[10] Angus Dempster, François Petitjean, and Geoffrey I Webb. 2020. ROCKET: exceptionally fast and accurate time series classification using random convolutional kernels. *Data Mining and Knowledge Discovery* 34, 5 (2020), 1454–1495.

[11] Cassandra A Dove, Fiona F Bradley, and Siddharth V Patwardhan. 2016. Seaweed biopolymers as additives for unfired clay bricks. *Materials and Structures* 49 (2016), 4463–4482.

[12] Michael Flynn, James Large, and Tony Bagnall. 2019. The contract random interval spectral ensemble (c-RISE): the effect of contracting a classifier on accuracy. In *Hybrid Artificial Intelligent Systems: 14th International Conference, HAIS 2019, León, Spain, September 4–6, 2019, Proceedings 14*. Springer, 381–392.

[13] Branko Glisic and Daniele Inaudi. 2007. *Fibre optic methods for structural health monitoring*. John Wiley & Sons.

[14] Mohamad Pauzi Ismail. 2017. Selection of suitable NDT methods for building inspection. In *IOP Conference Series: Materials Science and Engineering*, Vol. 271. IOP Publishing, 012085.

[15] Sergey Ivashov, Vladimir Razevig, and Andrey Zhuravlev. 2023. Application of microwave holographic subsurface technology for non-destructive testing of reinforced polyurethane foam. *Innovations in Aerospace Science and Technology* (2023), 001–004.

[16] Cun Ji, Mingsen Du, Yupeng Hu, Shijun Liu, Li Pan, and Xiangwei Zheng. 2022. Time series classification based on temporal features. *Applied Soft Computing* 128 (2022), 109494.

[17] Hamid Reza Khatami and Brendan C O'Kelly. 2013. Improving mechanical properties of sand using biopolymers. *Journal of Geotechnical and Geoenvironmental Engineering* 139, 8 (2013), 1402–1406.

[18] Byunghyun Kim and Soojin Cho. 2018. Automated vision-based detection of cracks on concrete surfaces using a deep learning technique. *Sensors* 18, 10 (2018), 3452.

[19] Matti Kuittinen, Caya Zernicke, Simon Slabik, and Annette Hafner. 2023. How can carbon be stored in the built environment? A review of potential options. *Architectural science review* 66, 2 (2023), 91–107.

[20] Hong-Cin Liou, Fabrizio Sabba, Aaron I Packman, George Wells, and Oluwaseyi Balogun. 2019. Nondestructive characterization of soft materials and biofilms by measurement of guided elastic wave propagation using optical coherence elastography. *Soft Matter* 15, 4 (2019), 575–586.

[21] Michel Serge Lorrain, Mônica Pinto Barbosa, Fº Silva, et al. 2011. Estimation of compressive strength based on Pull-Out bond test results for on-site concrete quality control. *Revista IBRACON de Estruturas e Materiais* 4 (2011), 582–591.

[22] S Mastrolitti, E Borsella, A Giuliano, MT Petrone, I De Bari, RJA Gosselink, G van Erven, E Annevelink, KS Traintafyllidis, and H Stichnothe. 2021. *Sustainable lignin valorization: Technical lignin, processes and market development.* Technical Report. IEA Bioenergy Task 42 & LignoCOST.

[23] DM McCann and MC Forde. 2001. Review of NDT methods in the assessment of concrete and masonry structures. *Ndt & E International* 34, 2 (2001), 71–84.

[24] Barney H Miao, Robert Headrick, Zhiye Li, Leonardo Spanu, David J Loftus, and Michael D Lepech. 2024. Life cycle assessment and design of LignoBlock: A lignin bound block on the path towards a green transition of the construction industry. *Journal of Cleaner Production* (2024), 143610.

[25] Matthew Middlehurst, Ali Ismail-Fawaz, Antoine Guillaume, Christopher Holder, David Guijo Rubio, Guzal Bulatova, Leonidas Tsaprounis, Lukasz Mentel, Martin Walter, Patrick Schäfer, and Anthony Bagnall. 2024. aeon: a Python toolkit for learning from time series. arXiv:2406.14231 [cs.LG] https://arxiv.org/abs/2406.14231

[26] Matthew Middlehurst, James Large, Michael Flynn, Jason Lines, Aaron Bostrom, and Anthony Bagnall. 2021. HIVE-COTE 2.0: a new meta ensemble for time series classification. *Machine Learning* 110, 11 (2021), 3211–3243.

[27] Sravan Muguda, Samuel John Booth, Paul Neil Hughes, Charles Edward Augarde, Celine Perlot, Agostino Walter Bruno, and Domenico Gallipoli. 2017. Mechanical properties of biopolymer-stabilised soil-based construction materials. *Géotechnique letters* 7, 4 (2017), 309–314.

[28] Ji-Sun Mun, Justin Alfred Pe III, and Sung-Phil Mun. 2021. Chemical characterization of kraft lignin prepared from mixed hardwoods. *Molecules* 26, 16 (2021), 4861.

[29] Leonard John Murdock. 1953. THE CONTROL OF CONCRETE QUALITY. *Proceedings of the Institution of Civil Engineers* 2, 4 (1953), 426–453.

[30] Christopher Niezrecki, Javad Baqersad, and Alessandro Sabato. 2018. Digital image correlation techniques for non-destructive evaluation and structural health monitoring. *Handbook of advanced non-destructive evaluation* 46 (2018).

[31] Francisco J Pallarés, Michele Betti, Gianni Bartoli, and Luis Pallarés. 2021. Structural health monitoring (SHM) and Nondestructive testing (NDT) of slender masonry structures: A practical review. *Construction and Building Materials* 297 (2021), 123768.

[32] Arti Patel, Marshall Thompson, Eric Harm, and William Sheftick. 1997. Developing QC/QA specifications for hot mix asphalt concrete in Illinois. *Transportation Research Record* 1575, 1 (1997), 66–74.

[33] JCM Pires. 2019. Negative emissions technologies: a complementary solution for climate change mitigation. *Science of the Total Environment* 672 (2019), 502–514.

[34] Amir Poursaee and Brandon Ross. 2022. The role of cracks in chloride-induced corrosion of carbon steel in concrete. *Corrosion and Materials Degradation* 3, 2 (2022), 258–269.

[35] Mohsin U Qureshi, Ilhan Chang, and Khaloud Al-Sadarani. 2017. Strength and durability characteristics of biopolymer-treated desert sand. *Geomech. Eng* 12, 5 (2017), 785–801.

[36] Henning Roedel, Isamar Rosa Plata, Michael Lepech, and David Loftus. 2015. Sustainability assessment of protein-soil composite materials for limited resource environments. *Journal of Renewable Materials* 3, 3 (2015), 183–194.

[37] Isamar Rosa, Henning Roedel, Maria I Allende, Michael D Lepech, and David J Loftus. 2020. On designing biopolymer-bound soil composites (BSC) for peak compressive strength. *Journal of Renewable Materials* 8, 8 (2020), 845–861.

[38] Mahdi Shariati, Nor Hafizah Ramli-Sulong, Mohammad Mehdi Arabnejad, Payam Shafigh, and Hamid Sinaei. 2011. Assessing the strength of reinforced concrete structures through Ultrasonic Pulse Velocity and Schmidt Rebound Hammer tests. *scientific research and essays* 6, 1 (2011), 213–220.

[39] KD Stanish, R Doug Hooton, Michael DA Thomas, et al. 2001. Testing the chloride penetration resistance of concrete: a literature review. (2001).

[40] Bing Wang, Shuncong Zhong, Tung-Lik Lee, Kevin S Fancey, and Jiawei Mi. 2020. Non-destructive testing and evaluation of composite materials/structures: A state-of-the-art review. *Advances in mechanical engineering* 12, 4 (2020), 1687814020913761.

[41] Jun Wang, Weizhuo Sun, Wenchi Shou, Xiangyu Wang, Changzhi Wu, Heap-Yih Chong, Yan Liu, and Cenfei Sun. 2015. Integrating BIM and LiDAR for real-time construction quality control. *Journal of Intelligent & Robotic Systems* 79 (2015), 417–432.

[42] Martine Wevers, Bart Nicolaï, Pieter Verboven, Rudy Swennen, Staf Roels, Els Verstrynge, Stepan Lomov, Greet Kerckhofs, Bart Van Meerbeek, Athina M Mavridou, et al. 2018. Applications of CT for non-destructive testing and materials characterization. *Industrial X-ray computed tomography* (2018), 267–331.

[43] Yulong Zhao, Jiaolong Ren, Ke Zhang, Yaofei Luo, and Kun Wang. 2024. Construction Quality Control for Rutting Resistance of Asphalt Pavement Using BIM Technology. *Buildings* 14, 1 (2024), 239.